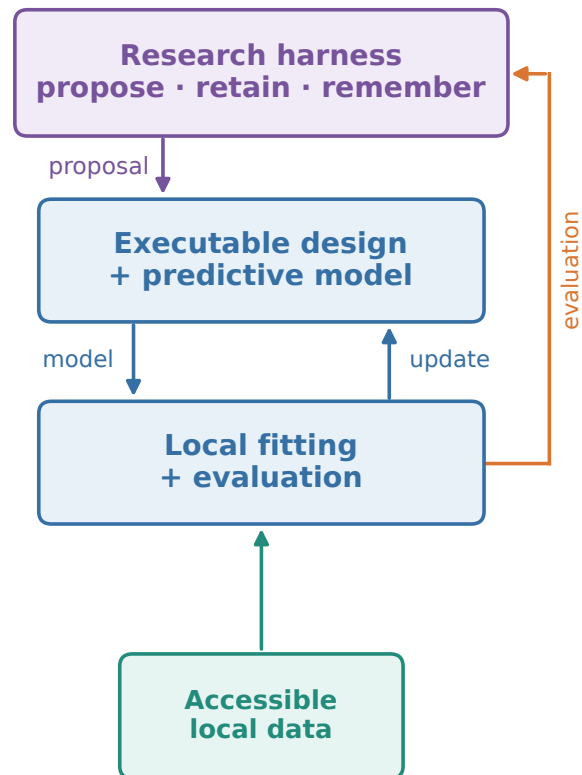


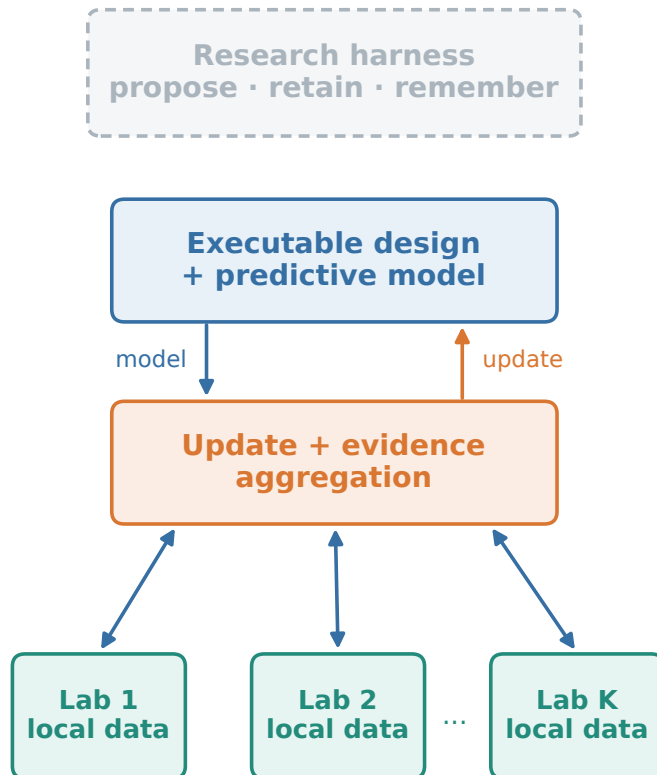
(a) Centralized bio-agent

one accessible dataset



(b) Collaborative training

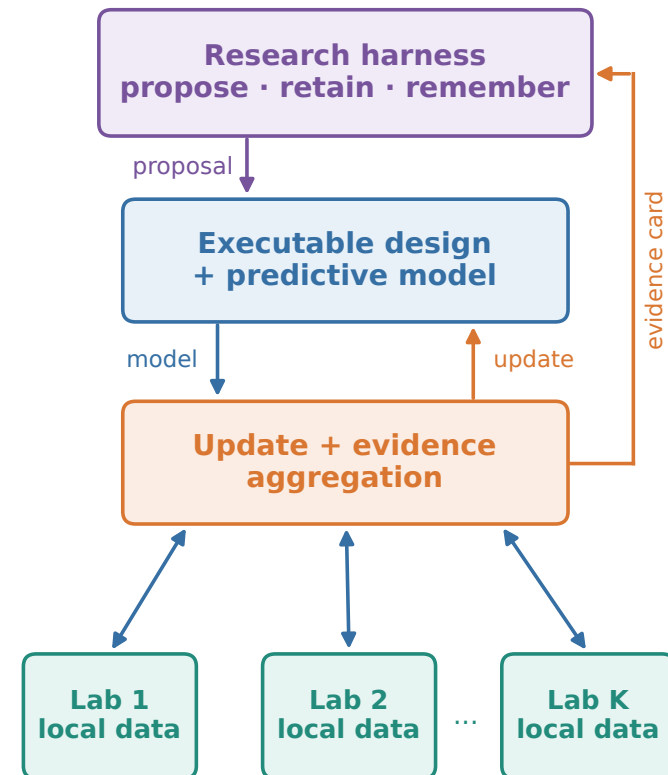
fixed executable design



Local fitting; updates + metrics shared

(c) BioCoLoop

model + design learning



Local fitting; updates + metrics shared

 local data

 predictive model

 research harness

 aggregation / evidence gate

Dashed gray: inactive research harness. K denotes participating laboratories.